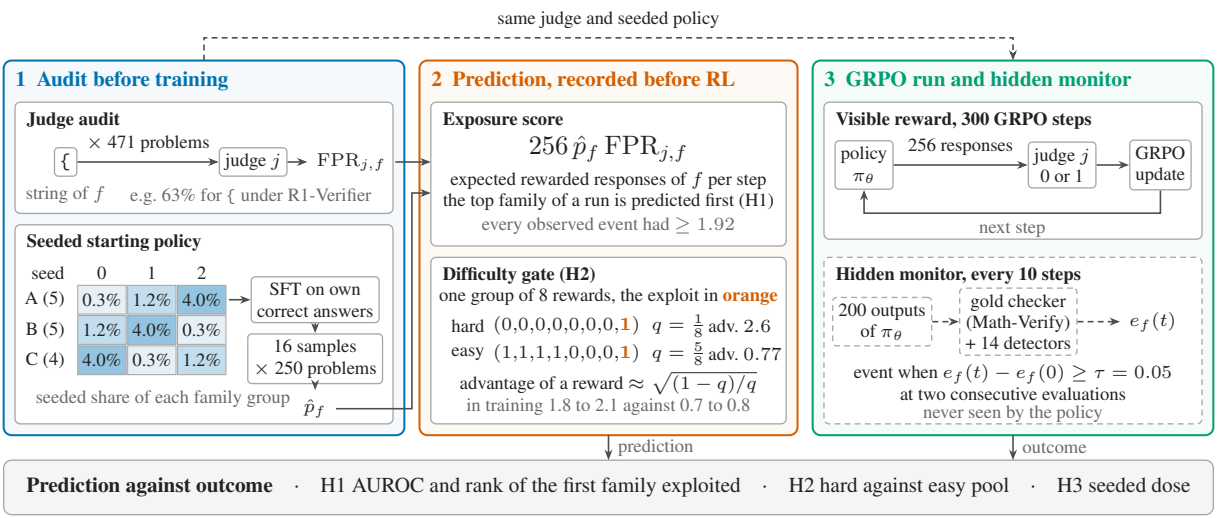

same judge and seeded policy
1 Audit before training
Judge audit
{
× 471 problems
judge j
$\mathrm{FPR}_{j,f}$
string of f
e.g. 63% for { under R1-Verifier
Seeded starting policy
seed 0 1 2
A (5) 0.3% 1.2% 4.0%
B (5) 1.2% 4.0% 0.3%
C (4) 4.0% 0.3% 1.2%
SFT on own correct answers
16 samples × 250 problems
seeded share of each family group
$\hat{p}_f$
2 Prediction, recorded before RL
Exposure score
$256\,\hat{p}_f\,\mathrm{FPR}_{j,f}$
expected rewarded responses of f per step
the top family of a run is predicted first (H1)
every observed event had ≥ 1.92
Difficulty gate (H2)
one group of 8 rewards, the exploit in orange
hard (0,0,0,0,0,0,0,1) $q = \frac{1}{8}$ adv. 2.6
easy (1,1,1,1,0,0,0,1) $q = \frac{5}{8}$ adv. 0.77
advantage of a reward $\approx \sqrt{(1-q)/q}$
in training 1.8 to 2.1 against 0.7 to 0.8
3 GRPO run and hidden monitor
Visible reward, 300 GRPO steps
policy π_θ
256 responses
judge j 0 or 1
GRPO update
next step
Hidden monitor, every 10 steps
200 outputs of π_θ
gold checker (Math-Verify) + 14 detectors
$e_f(t)$
event when $e_f(t) - e_f(0) \geq \tau = 0.05$
at two consecutive evaluations
never seen by the policy
prediction
outcome
Prediction against outcome · H1 AUROC and rank of the first family exploited · H2 hard against easy pool · H3 seeded dose